

Explaining Variability in Shell-Based Oxygen Isotope Ratios in the Western North Atlantic Using a Proxy System Model

Soraya S. Remaili¹, Nina M. Whitney^{2,3}, Branwen Williams⁴, Caroline C. Ummenhofer³, Arne Biastoch⁵, Torge Martin⁵, Alan D. Wanamaker⁶, Diana L. Thatcher⁶, James Berg⁷, Amanda Hauser⁶, Heeyeon Sun^{6,8}, Alex Walton⁶, Madelyn J. Mette⁹, Eric Powell¹⁰, Roger Mann¹¹, Eileen Hoffman¹², John Klinck¹².

¹University of South Carolina, Columbia, ²Western Washington University, ³Woods Hole Oceanographic Institution, ⁴Claremont McKenna College, ⁵GEOMAR Helmholtz Centre for Ocean Research, ⁶Iowa State University, ⁷University of New South Wales, ⁸University of Massachusetts, Amherst, ⁹U.S.G.S., St. Petersburg Coastal and Marine Science Center, ¹⁰University of Southern Mississippi, ¹¹Virginia Institute of Marine Science, ¹²Old Dominion University

I. INTRODUCTION

- Over the last 100 years, the Gulf of Maine has undergone substantial changes in sea surface temperature, with the area warming more rapidly than the rest of the global oceans since 1980 (Pershing et al., 2015).
- To investigate whether this warming trend is a recent phenomenon or a long-term signal, it is necessary to use paleoclimate proxies to look back further than modern instrumentation.
- Marine calcifiers, like corals and bivalves, can document seawater conditions via the oxygen isotopic signature of their carbonate shells (Williams et al., 2024), and by measuring the ratio of oxygen isotopes relative to the seawater in the shell of a clam, the temperature of the water that the clam was living in can be estimated.
- Here, we investigate previously developed oxygen isotope records from six different sites (Fig. 1). When looking at these six clam sampling sites, large amounts of variability can be seen between the oxygen isotopic records (Fig. 2).
- This variability between sites might be due to one site's oxygen isotope record might being more influenced by seawater oxygen isotope variability while another site's is more influenced by temperature variability.
- This variability is investigated further using a strategy called proxy system modeling (Evans et al., 2013), where a proxy system model, or a PSM, is built to evaluate the strength of a proxy's signal and better understand the effect of different processes on the proxy's signal.

III. RESULTS

- Out the 360 models constructed, 49 of the models were well-modeled (Fig. 5), that is, had both a significant p-value and a positive correlation coefficient, and were examined further at the level of each site (Table 1).

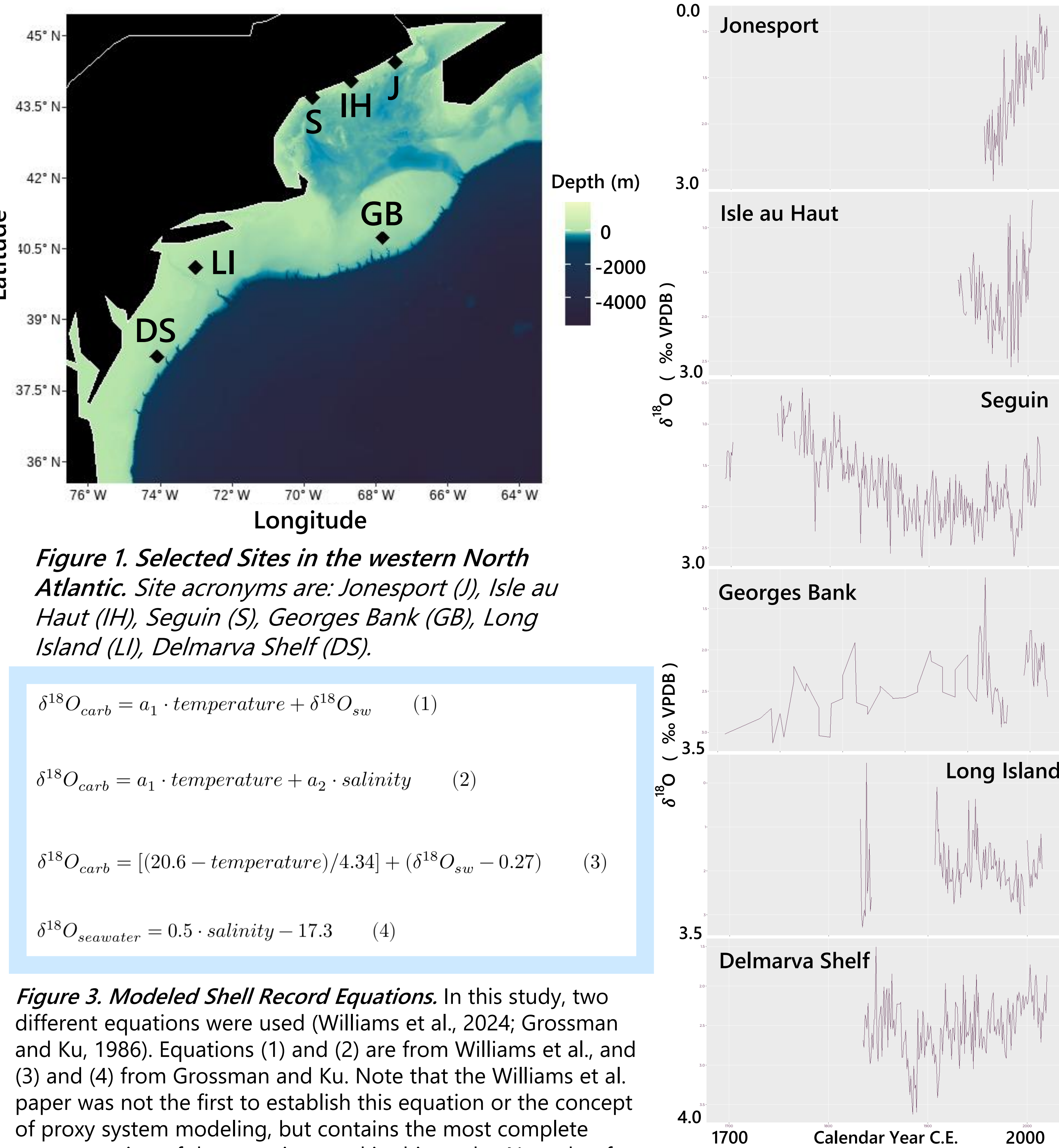
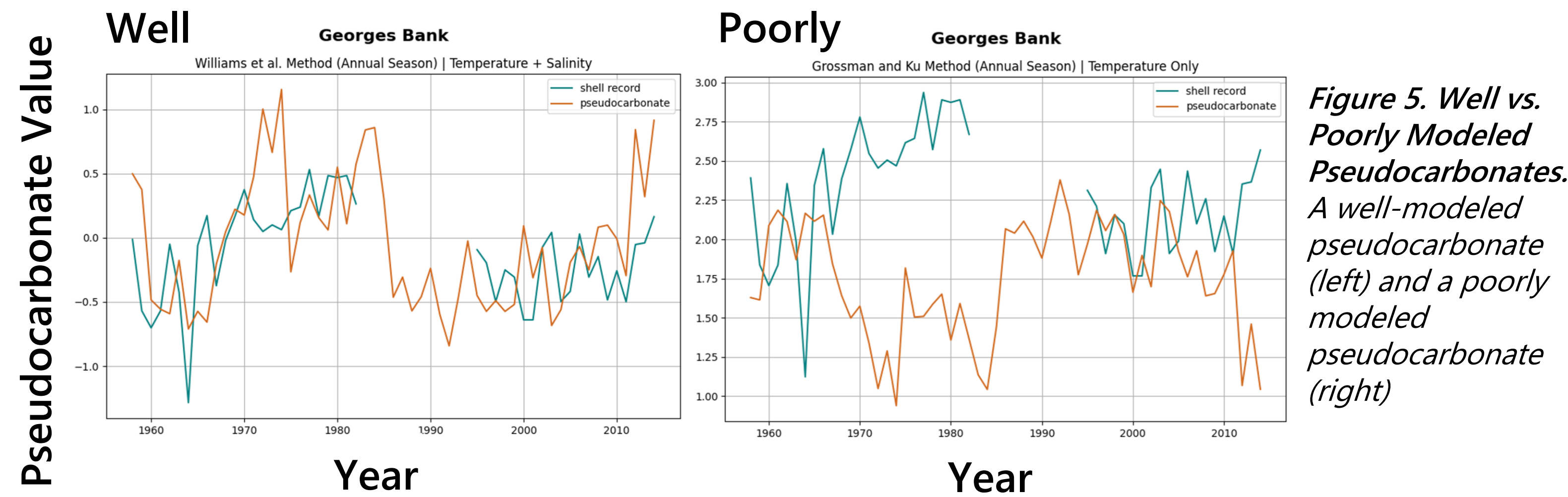


Figure 3. Modeled Shell Record Equations. In this study, two different equations were used (Williams et al., 2024; Grossman and Ku, 1986). Equations (1) and (2) are from Williams et al., and (3) and (4) from Grossman and Ku. Note that the Williams et al. paper was not the first to establish this equation or the concept of proxy system modeling, but contains the most complete representation of the equation used in this study. Note that for this region, $a_1 = 0.22$ and $a_2 = 0.55$.

Figure 6. Average Yearly Salinity for VIKING20X. For VIKING20X, variance in salinity between sites is significantly less than that of other products, particularly iCESM. SODA and GLORYS perform similarly and are more varied across time.

Figure 7. Average Yearly Temperature for VIKING20X. Temperature varies in a similar way to salinity, but the three northernmost sites differ less temperature-wise than they do salinity-wise.

IV. CONCLUSIONS

- A diverse array of combinations of different model inputs, products, and equations produced well-modeled pseudocarbonates at each of the six different sites.
- The results of this study suggest that the variability of the different oxygen isotope records constructed in this region is either primarily temperature-driven, salinity-driven, or driven by a combination of both temperature and salinity. Differences in dominant environmental variability drivers between sites may help to explain instances of low correlation between oxygen isotope records in this region.
- Additionally, the results of this study also suggest that the Grossman and Ku equation suited the bivalve pseudocarbonates more than the Williams et al. equation, increasing modeling capability of the equation historically used for bivalves (Grossman and Ku) versus that used for tropical corals (Williams et al.)
- Also, though the results can suggest that certain model conditions are better at representing isotope record variability, further and more thorough study is needed to investigate the role of these model factors more and weigh their role with the oxygen isotope record itself, as well as investigate the role of the mixed results given by models (Fig. 6 and 7) vs. the bivalve records themselves.